

FEATURE ENGINEERING USING PYTHON

Internship Project Report

Submitted By

Shwetha Kalagi

Organization

RD INFRO TECHNOLOGY

Domain

Artificial Intelligence

Task

Task 3 – Feature Engineering

1. INTRODUCTION

Feature Engineering is the process of transforming raw data into useful features that improve the performance of Artificial Intelligence and Machine Learning models. It is an important step in data preprocessing and helps prepare datasets for model training and analysis.

In this project, feature engineering techniques were applied to an employee dataset using Python and Scikit-Learn.

2. OBJECTIVE

- Convert categorical data into numerical format.
 - Apply encoding techniques to categorical features.
 - Scale numerical data using StandardScaler.
 - Prepare data for Artificial Intelligence and Machine Learning applications.
-

3. TOOLS USED

- Python
- Pandas
- Scikit-Learn
- Google Colab

4. LIBRARIES USED

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

5. DATASET DESCRIPTION

The dataset contains employee information including Name, Department, and Salary.

Name	Department	Salary
Rahul	IT	35000
Priya	HR	45000
Amit	Finance	50000
Sneha	Marketing	41000
Karan	IT	40000

6. METHODOLOGY

Step 1: Label Encoding

The Department column was converted into numerical values using Label Encoding.

Step 2: One-Hot Encoding

One-Hot Encoding was applied to create separate binary columns for each department category.

Step 3: Feature Scaling

StandardScaler was used to standardize salary values so that they can be used effectively in machine learning models.

7. SOURCE CODE

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler

data = {
```

```

    'Name': ['Rahul', 'Priya', 'Amit', 'Sneha', 'Karan'],
    'Department': ['IT', 'HR', 'Finance', 'Marketing', 'IT'],
    'Salary': [35000, 45000, 50000, 41000, 40000]
}

df = pd.DataFrame(data)

print("Original Dataset")
print(df)

label_encoder = LabelEncoder()
df['Department_Label'] = label_encoder.fit_transform(df['Department'])

print("\nAfter Label Encoding")
print(df)

one_hot = pd.get_dummies(df['Department'], prefix='Dept')

print("\nOne Hot Encoded Data")
print(one_hot)

scaler = StandardScaler()
df['Salary_Scaled'] = scaler.fit_transform(df[['Salary']])

print("\nFinal Feature Engineered Dataset")
print(df)

```

8. OUTPUT

Original Dataset

	Name	Department	Salary
0	Rahul	IT	35000
1	Priya	HR	45000
2	Amit	Finance	50000
3	Sneha	Marketing	41000
4	Karan	IT	40000

After Label Encoding

	Name	Department	Salary	Department_Label
0	Rahul	IT	35000	2
1	Priya	HR	45000	1
2	Amit	Finance	50000	0
3	Sneha	Marketing	41000	3
4	Karan	IT	40000	2

One-Hot Encoded Data

	Dept_Finance	Dept_HR	Dept_IT	Dept_Marketing
0	False	False	True	False
1	False	True	False	False
2	True	False	False	False
3	False	False	False	True
4	False	False	True	False

Final Feature Engineered Dataset

	Name	Department	Salary	Department_Label	Salary_Scaled
0	Rahul	IT	35000	2	-1.429743
1	Priya	HR	45000	1	0.556011
2	Amit	Finance	50000	0	1.548888
3	Sneha	Marketing	41000	3	-0.238290
4	Karan	IT	40000	2	-0.436866

9. RESULTS

The employee dataset was successfully transformed using feature engineering techniques. Label Encoding converted categorical data into numerical values, One-Hot Encoding created binary features, and StandardScaler normalized salary values. The final dataset is suitable for Artificial Intelligence and Machine Learning applications.

10. CONCLUSION

This project demonstrates the implementation of Feature Engineering using Python and Scikit-Learn. The dataset was successfully preprocessed using Label Encoding, One-Hot Encoding, and Feature Scaling techniques. These methods improve data quality and prepare datasets for Artificial Intelligence and Machine Learning models.